

Run 04 | TEA

India

IV only

Target Total early-stage entrepreneurial activity; binary 1 vs. 0
Sample 27,555 respondents **Coverage** 1 country; 2015–2022
Positive cases 3,188 (11.57%) **Case time** 50s
Train / val. / test 18,739 / 3,306 / 5,510 (68% / 12% / 20%, stratified)
Predictors (13): Self-efficacy, opportunity perception, fear of failure, knowing entrepreneurs, intention, media coverage, business exit, business angel, established business owner, good career choice, ease of starting a business, high status, social enterprise.
Legatum predictors (13): Prosperity index score, Safety and security, Personal freedom, Governance, Social capital, Investment environment, Enterprise conditions, Market access and infrastructure, Economic quality, Living conditions, Health, Education, Natural environment. Total independent variables: 26.
Controls (0): None. Country and respondent ID are excluded as predictors.
Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: CatBoost (0.9974). Highest F1: H2O GBM (0.9363). The all-negative accuracy baseline is 88.43% (full sample).

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
CatBoost	0.9974	0.9775	98.42	91.29	95.45	0.9332	10s
XGBoost	0.9971	0.9749	98.37	91.63	94.51	0.9304	<1s
H2O GBM	0.9971	0.9737	98.48	90.60	96.86	0.9363	3s
Elastic Net	0.9969	0.9711	98.37	89.70	97.02	0.9321	9s
Random Forest	0.9969	0.9724	98.46	92.07	94.82	0.9343	<1s
Logistic Regression	0.9969	0.9709	98.40	90.31	96.55	0.9332	<1s
Lasso	0.9968	0.9706	98.31	91.09	94.66	0.9284	6s
Ridge	0.9968	0.9693	98.37	90.40	96.08	0.9315	12s
Decision Tree	0.9941	0.9209	98.46	91.57	95.45	0.9347	<1s
Naive Bayes	0.9349	0.6756	93.09	66.33	81.63	0.7319	6s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 8

Rank	Predictor	Raw importance	Share%
1	Entrepreneurial intention	1,730.97	59.72
2	Established business owner	527.92	18.21
3	Knowing entrepreneurs	73.76	2.54
4	Business angel	71.36	2.46
5	Business exit	52.89	1.82
6	Self-efficacy	44.43	1.53
7	Business exit (missing indicator)	30.05	1.04
8	Opportunity perception	24.70	0.85

Share is the percentage of total forest impurity importance across all 51 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Leading Legatum predictors within the same forest

Rank	Legatum predictor	Raw importance	Share%
1	Prosperity index score	15.83	0.55
2	Investment environment	14.94	0.52
3	Safety and security	13.20	0.46

Legatum subset ranked by raw importance; shares use total forest importance, not the Legatum subtotal.

Source: Completed Objective 2 batch, 28 September 2026; 04_obj2_india_iv_TEA. Scores and importance are taken directly from saved CSV outputs.